Belinda Z. Li

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Education

Massachusetts Institute of Technology 2020 –

PhD student, Electrical Engineering and Computer Science

- Advisor: Jacob Andreas

University of Washington 2015 – 2019

Bachelors of Science, Computer Science, Magna Cum Laude

- Cumulative GPA: 3.95 / 4.0, Major GPA: 3.96 / 4.0
- Highlighted Coursework: Natural Language Processing, NLP Capstone, Machine Learning, Artificial Intelligence, Algorithms, Foundations of Computing I/II (Discrete Math & Probability), Honors Accelerated Calculus I/II/III
- Advisor: Luke Zettlemoyer

Peer-Reviewed Publications

[1] Nayeon Lee, **Belinda Z. Li**, Sinong Wang, Wen-tau Yih, Hao Ma, and Madian Khabsa. 2020. Language models as fact checkers? In *Proceedings of the Third Workshop on Fact Extraction and VERification (FEVER)*, pages 36–41. Association for Computational Linguistics

[2] **Belinda Z. Li**, Gabriel Stanovsky, and Luke Zettlemoyer. 2020. Active learning for coreference resolution using discrete annotation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8320–8331. Association for Computational Linguistics

Preprints

[A] Sinong Wang, **Belinda Z. Li**, Madian Khabsa, Han Fang, and Hao Ma. 2020. Linformer: Self-attention with linear complexity

Industry/Research Experience

Facebook AI Aug. 2019 – Sep. 2020

Software Engineer, AI Integrity Team (Facebook AI Applied Research)

- (Product) Built multimodal representations for Facebook posts pretrained with self-supervision. Shipped to hate speech domain, improving offline hate speech prevalence metrics by 25%
  - Work featured in part of this post: https://ai.facebook.com/blog/ai-advances-to-better-detect-hate-speech/
- (Research) Worked on Linformer [A]: discover that the self-attention mechanism in Transformers is low rank, and exploit this to create linear-time transformers, greatly speeding up both training and inference on long sequences
- (Research) Devised models for fake news detection: using multi-tasking, or leveraging implicit knowledge in pretrained language models [1]
- (Research) Devised an efficient entity linking model for questions

University of Washington Jan. 2018 – June 2019

Undergraduate Research Assistant, with Luke Zettlemoyer

- Devised new annotation technique for active learning for coreference resolution, see [2]
- Built models for entity-to-entity sentiment analysis

Facebook June 2018 – Aug. 2018

Software Engineer Intern, Bot Detection Team

- Worked on front-end script for tracking and storing user mouse movements
- Classified mouse movements for bot detection, using heuristic techniques, as well as simple ML models (gradient boosting tree classifiers, K-means clustering)

Teaching

University of Washington Sep. 2017 – June 2019

Teaching Assistant

- Responsibilities (mostly) entailed grading, leading weekly section, hosting office hours and review sessions, and weekly meetings with course staff. List of courses below:

CSE 311: Foundations of Computing I (Discrete Math), *with Kevin Zatloukal*	Spring 2019
CSE 312: Foundations of Computing II (Probability/Statistics), *with Martin Tompa*	Winter 2019
CSE 312: Foundations of Computing II (Probability/Statistics), *with Martin Tompa*	Winter 2018
CSE 331: Software Design and Implementation, *with Kevin Zatloukal*	Autumn 2017

Awards

Ida M. Green Memorial Fellowship 2020 – 2021

Denice Dee Denton Scholars Endowment 2018 – 2019

Dean's List 2015 – 2017 (annual), SP18/F18/SP19 (quarterly)

National Merit Scholar 2016

Service

Program Committee: COLING 2020, EMNLP 2020

Student Volunteer: ACL 2020